

Problem Statement

There is a right-angled triangle ABC ($\angle A = 90^\circ$). On the same plane, outside $\triangle ABC$, construct isosceles triangles PBC , QCA , and RAB such that sides BC , CA , and AB are their respective bases, and each has a vertex angle of 120° . Show that $\triangle PQR$ is an equilateral triangle.

0.1 Rigorous Algebraic Proof

Proof. We can model this configuration on the complex plane to track the rotational shifts efficiently. By placing the right-angled vertex A directly at the origin, we establish a clean frame of reference:

$$A = 0$$

Let vertex B extend a real length c horizontally, while vertex C extends an imaginary height b vertically:

$$B = c, \quad C = ib$$

Evaluating the outward construction $\triangle QCA$, the target vertex shape contains a vertex angle of $\angle Q = 120^\circ$. This assigns baseline angles of $\angle QCA = \angle QAC = 30^\circ$. From a transform perspective, vertex Q is located by taking the vector C , rotating it clockwise by 30° , and scaling it by the factor $\frac{1}{\sqrt{3}}$. Using exponential complex markers ($\omega = e^{i\pi/6}$):

$$Q = \frac{1}{\sqrt{3}} C e^{-i\pi/6} = \frac{1}{\sqrt{3}} (ib) \left(\frac{\sqrt{3}}{2} - \frac{1}{2}i \right) = \frac{b}{2\sqrt{3}} + i\frac{b}{2}$$

Following the identical mapping sequence for the outward form $\triangle RAB$ built over the base segment AB :

$$R = \frac{1}{\sqrt{3}} B e^{i\pi/6} = \frac{1}{\sqrt{3}} (c) \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i \right) = \frac{c}{2} + i\frac{c}{2\sqrt{3}}$$

To map the final peak position P built onto the hypotenuse base BC , we apply a 30° clockwise rotation to the vector pointing from B to C , scaling its length by $\frac{1}{\sqrt{3}}$:

$$P - B = \frac{1}{\sqrt{3}} (C - B) e^{-i\pi/6}$$

$$P = c + \frac{1}{\sqrt{3}} (ib - c) \left(\frac{\sqrt{3}}{2} - \frac{1}{2}i \right) = \left(\frac{c}{2} + \frac{b}{2\sqrt{3}} \right) + i \left(\frac{b}{2} + \frac{c}{2\sqrt{3}} \right)$$

To demonstrate that $\triangle PQR$ is equilateral, it suffices to prove that a 60° complex rotation vector ($e^{i\pi/3}$) maps the side vector $\vec{RQ}$ directly onto $\vec{RP}$:

$$\text{Condition: } (Q - R) e^{i\pi/3} = P - R$$

Let us first discover the individual structural vectors from our coordinates:

$$Q - R = \left(\frac{b}{2\sqrt{3}} - \frac{c}{2} \right) + i \left(\frac{b}{2} - \frac{c}{2\sqrt{3}} \right)$$

$$P - R = \frac{b}{2\sqrt{3}} + i \frac{b}{2}$$

Now, multiplying our vector $(Q - R)$ by the rotation constant $e^{i\pi/3} = \frac{1}{2} + i\frac{\sqrt{3}}{2}$:

$$(Q - R)e^{i\pi/3} = \left[\left(\frac{b}{2\sqrt{3}} - \frac{c}{2} \right) + i \left(\frac{b}{2} - \frac{c}{2\sqrt{3}} \right) \right] \left[\frac{1}{2} + i\frac{\sqrt{3}}{2} \right]$$

Expanding the real distribution terms:

$$\text{Real} = \frac{1}{2} \left(\frac{b}{2\sqrt{3}} - \frac{c}{2} \right) - \frac{\sqrt{3}}{2} \left(\frac{b}{2} - \frac{c}{2\sqrt{3}} \right) = \frac{b}{4\sqrt{3}} - \frac{c}{4} - \frac{b\sqrt{3}}{4} + \frac{c}{4} = \frac{b}{2\sqrt{3}}$$

Expanding the imaginary distribution components:

$$\text{Imaginary} = \frac{\sqrt{3}}{2} \left(\frac{b}{2\sqrt{3}} - \frac{c}{2} \right) + \frac{1}{2} \left(\frac{b}{2} - \frac{c}{2\sqrt{3}} \right) = \frac{b}{4} - \frac{c\sqrt{3}}{4} + \frac{b}{4} - \frac{c}{4\sqrt{3}} = \frac{b}{2}$$

Combining these evaluated coordinates confirms the transformation matches perfectly:

$$(Q - R)e^{i\pi/3} = \frac{b}{2\sqrt{3}} + i \frac{b}{2} = P - R$$

Since the rotation transforms the side segments completely in length and orientation by exactly 60° , the resulting structure forms a verified equilateral shape. $\square$